BARRIER SERIES (Barrier-MA1 / Barrier-MK2)

InsulTherm Barrier™

4-Layer 1,100°C Nanoporous Aerogel-Mica Thermal Runaway & Fire Containment Sheet

InsulTherm Barrier™ is an ultra-thin, 4-layer inter-cell thermal runaway firewall engineered for utility-scale LiFePO4 Battery Energy Storage Systems (BESS) and high-density NMC/LFP electric vehicle traction packs. Combining a 0.20 mm Phlogopite mica mineral blast shield with a 1.20 mm nanoporous silica aerogel core (k = 0.018 W/m·K), Barrier™ blocks both 1,100°C particle-laden venting torches and conductive heat transfer between adjacent prismatic or pouch cells.



Macro Cross-Section / Studio Hardware Inspection

k = 0.018 W/m·K	**>1,100°C (>5 min)**	**ARAI AIS-156**	**UL 9540A Support**	**>15 kV/mm**

1. CHEMICAL / MATERIAL COMPOSITION & INTERNAL LAYER ARCHITECTURE

Hybrid mineral-nanoporous composite stack: (1) Calcined Phlogopite mica paper impregnated with high-temp silicone resin to resist abrasive hot-particle ejection during cell vent bursts; (2) Fiber-reinforced mesoporous amorphous silica aerogel felt (pore size ~20 nm, suppressing gas-phase convection); (3) 0.10 mm flame-retardant acrylic PSA; and (4) Siliconized kraft/PET liner.

Layer 1 (0.20 mm)	Phlogopite Mica Flame Shield™ Blast Face (>1,100°C Torch & Particle Guard)
Layer 2 (1.20 mm)	Nanoporous Silica Aerogel Felt Core™ (k = 0.018 W/m·K Mesoporous Block)
Layer 3 (0.10 mm)	High-Temperature Flame-Retardant Acrylic PSA Mounting Adhesive
Layer 4 (Liner)	Siliconized Kraft Paper / PET Differential Release Liner (CNC Kiss-Cut)

2. TYPICAL PHYSICAL, THERMAL & ELECTRICAL PROPERTIES

PROPERTY / PARAMETER	TYPICAL ENGINEERING VALUE	TEST METHOD / STANDARD
Thermal Conductivity @ 25°C	**0.018 W/m·K (0.023 W/m·K @ 200°C)**	ASTM C518 / ASTM D5470
Direct Flame Hold Endurance	**> 1,100°C torch for > 5 mins (Zero burn-through)**	Internal Torch / UL 9540A
ARAI AIS-156 Pack FEA Benchmark	**Adjacent cell at 74.189°C (<80°C cascade limit)**	4-Quadrant Pack Validation
Dielectric Breakdown Strength	**> 15 kV/mm (MA1) / > 22 kV/mm (MK2)**	ASTM D149
Flammability Classification	**UL 94 V-0 / Non-propagating mineral-aerogel**	UL 94 / IEC 60695
Nominal Composite Thickness	**1.50 mm std (1.0 mm – 3.0 mm custom)**	ASTM D374
Continuous / Excursion Temp	**-50°C to +650°C cont. / >1,100°C excursion**	TGA Thermal Analysis
Compression Deflection	**10% – 25% deflection @ 50 – 150 kPa**	ASTM D575
Bulk Composite Density	**0.22 – 0.28 g/cm³ (Low EV/Aero mass impact)**	ASTM C303
Particulate / Dust Control	**Mica-faced & edge-consolidated assembly**	InsulTherm QC-BAR-02

InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India

Engineering Notice: Typical laboratory & baseline metrics (ASTM D5470 / UL 94 / ASTM E595). System-level OEM qualification is required prior to volume release.

3. ORDERING NOMENCLATURE, PREMIX VARIANTS & COMMERCIAL PRICING SCHEDULE

PART / GRADE CODE	MATERIAL SPECIFICATION & CONSTRUCTION	PROTOTYPE / R&D TIER	VOLUME / OEM TIER
Barrier-MA1-150	0.20mm Mica + 1.20mm Silica Aerogel + FR-PSA (100 cm² ref. cell sheet)	**₹110 / sheet (₹1.10/cm²)**	**₹85 / sheet (₹0.85/cm² area)**
Barrier-MK2-165	Dual-Mica + Aerogel + Kapton® HV Encapsulation (100 cm² ref. sheet)	**₹110 / sheet (₹1.10/cm²)**	**₹85 / sheet (₹0.85/cm² area)**
Custom BESS / AIS-156	280Ah/314Ah LFP Custom Cutouts (Exact quote by sheet cm²; Kit ₹14,500)	**₹1.10 / cm² (min ₹55/pc)**	**₹0.85 / cm² nested (≥100 pcs)**

4. TARGET INDUSTRIAL APPLICATIONS & DEPLOYMENT SECTORS

- Grid-scale 20-ft BESS containers: vertical inter-cell thermal runaway firewalls between 280Ah–314Ah prismatic LiFePO4 cells.
- EV 2W/3W/4W & commercial fleet battery modules engineered for ARAI AIS-156 Amendment 3 five-minute thermal propagation compliance.
- Battery enclosure lid flame-deflection blankets preventing hot gas/electrolyte jetting burn-through of aluminum or composite covers.
- Defense, marine, and aviation energy storage modules requiring high dielectric holdoff after severe thermal shock.

5. DIGITAL CNC CONVERTING, APPLICATION & STORAGE GUIDELINES

1. Orientation: Always orient the rigid Phlogopite Mica blast face toward the highest thermal-runaway impingement surface or vent path; apply the PSA face to clean, IPA-degreased cell walls or module end-plates. 2. Compression Preload: Design cell-stack preload between 35 kPa and 120 kPa at Beginning-of-Life (BOL) to absorb End-of-Life (EOL) swelling. 3. Regulatory Note: UL 9540A and ARAI AIS-156 evaluate system-level propagation; InsulTherm supplies nested kits and lot traceability for pack testing.

6. QUALITY ASSURANCE, REGULATORY COMPLIANCE & SOVEREIGN DVA TRACEABILITY

Thermal & Electrical Metrology	Batch-wise ASTM D5470-17 thermal impedance calibration curves (10–50 psi) & ASTM D149 dielectric verification.
Flammability & Fire Safety	UL 94 V-0 rated material formulations; supports ARAI AIS-156 Amendment 3 & UL 9540A pack/rack propagation testing.
Outgassing & Environmental	RoHS 3 (EU 2015/863) & REACH compliant; ASTM E595 low-outgassing trace on Graphite-Flex™ & Ultra-NS™ grades.
Domestic Value Addition (DVA)	100% India Digital CNC Converting (Gujarat); compliant with PM E-DRIVE, SECI BESS DVA & <₹200 Cr MoD GTE Ban.

RAPID 24–48 HOUR CAD-TO-PART PROTOTYPING & OEM PROCUREMENT DESK
Submit .DXF / .STEP drawings, request <₹15,000 Petty-Cash R&D Sampler Kits, or schedule an engineering review:
Primary Official Desk: manav.sb@inslthrm.in | **Active Operations & Order Routing:** manavsb9719@gmail.com
Engineering Helpline: +91 78148 XXXXX | **Manufacturing Wing:** InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India